THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · MEDIUM LEVEL UNIT TEST SET 2

Chapters 10–13: **Light: Mirrors and Lenses, Keeping Time with the Skies, How Nature Works in Harmony, Our Home: Earth, a Unique Life-sustaining Planet** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Venus has a surface temperature of about 460 degrees Celsius mainly due to: 1

(a) its magnetic field **(b)** a thick CO2 atmosphere
(c) closeness to the Moon **(d)** its oceans

2. A mirror is needed for the side of a car to show a wide area behind the driver. The best choice is a: 1

(a) cylindrical glass block **(b)** concave mirror
(c) plane mirror **(d)** convex mirror

3. The difference between the solar year (365 days) and the lunar year (354 days) is about: 1

(a) 5 days **(b)** 30 days **(c)** 1 day **(d)** 11 days

4. Burning of coal increases carbon dioxide and the effect on the atmosphere is: 1

(a) global warming **(b)** cooling **(c)** no effect **(d)** more oxygen

5. The inner core is solid even though it is very hot because of: 1

(a) no gravity **(b)** low temperature
(c) water **(d)** extremely high pressure

6. A concave lens forms an image of an object. The image is: 1

(a) real and larger **(b)** real and inverted
(c) virtual, erect and smaller **(d)** virtual and larger

7. The full Moon rises at about: 1

(a) sunset **(b)** noon **(c)** sunrise **(d)** midnight

8. A tiger population in a forest falls. The deer population will probably: 1

(a) remain zero **(b)** vanish **(c)** increase **(d)** decrease

9. In 200 L of air the volume of nitrogen is about: 1

(a) 42 L **(b)** 78 L **(c)** 156 L **(d)** 21 L

10. A man walks towards a plane mirror at 2 m/s. The speed of his image towards him is: 1

(a) 0 **(b)** 2 m/s **(c)** 4 m/s **(d)** 1 m/s

11. If today is a full moon, the next new moon will be after about: 1

(a) 60 days **(b)** 15 days **(c)** 7 days **(d)** 30 days

12. In a food chain, the amount of energy available is greatest at the: 1

(a) top carnivore level **(b)** secondary consumer level
(c) producer level **(d)** primary consumer level

13. The Earth's water is about 97 per cent salt water. If total water is 1000 units, the units of fresh water are about: 1

(a) 300 **(b)** 970 **(c)** 30 **(d)** 3

14. A concave mirror held close to the face shows an image that is: 1

(a) virtual, erect and enlarged **(b)** virtual and diminished
(c) real and inverted **(d)** real and diminished

15. After about 3 years, the lunar calendar falls behind the solar by about: 1

(a) 33 days **(b)** 1 day **(c)** 22 days **(d)** 11 days

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Why does the lunar year have 354 days whereas the solar year has 365 days? 2

17. Why does overuse of pesticides harm an ecosystem? 2

18. Why does the temperature fall with height in the troposphere but rise in the stratosphere? 2

19. Why does a convex lens form a real image of a distant object? 2

20. If the shadow of a tree is 6 m when a 2 m stick has a shadow of 1 m, find the height of the tree. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Explain why water exists in all three states on Earth but not on the Moon. 3

22. Explain what happens to a parallel beam of light incident on a convex mirror and on a concave mirror. 3

23. Why is the day longer in summer? Explain with the tilt of the Earth. 3

24. Explain how rainwater harvesting helps in conserving water. 3

OR

Explain why the sky appears blue during the day from the surface of the Earth. 3

25. Explain why we see a well-polished wooden table's reflection but not that of a rough one. 3

OR

Explain how the shadow of a stick can be used to find the north-south direction at noon. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. A man 1.8 m tall stands in front of a plane mirror. Explain the minimum size of mirror and where to place it to see his full image. Show the geometry in words. 5

OR

A traveller finds the Pole Star 25 degrees above the northern horizon. Find his latitude and describe how one can locate the Pole Star with the Big Dipper. 5

27. Compare a pond and a grassland ecosystem on the basis of producers, consumers, decomposers, abiotic factors, and give one food chain for each. 5

OR

Explain the causes and effects of ozone depletion, mention the gases responsible, and the steps taken by the world to repair the layer. 5

END OF QUESTION PAPER **Total: 50 Marks**

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